AI

Advanced Machine Learning MCQ Quiz

Regression • Overfitting • Underfitting • Generalization

Dr. Elfikky Active learning Task before he lecture

Artificial Intelligence

QUESTION 1

REGRESSION

A university wants to predict a student's final GPA from attendance, previous GPA, study hours, and assignment scores. Which learning problem is most appropriate?

A Binary classification

B Multi-class classification

C Regression

D Clustering

QUESTION 2

OVERFITTING

A regression model performs extremely well on its training data but produces large errors on new test data. What is the most likely problem?

A Underfitting

B Overfitting

C Clustering

D Dimensionality reduction

QUESTION 3

UNDERFITTING

Which situation best describes underfitting?

A The model memorizes the training examples but fails on new data.

B The model performs poorly on both training and testing data.

C The model performs poorly on training data but perfectly on testing data.

D The model has high accuracy only because the dataset is large.

Consider the following prediction problems. Which one is NOT a regression problem?

A Predicting tomorrow's temperature

B Predicting the selling price of a house

C Predicting whether an email is spam or not spam

D Predicting a student's exam score

In supervised learning, what is the main difference between a feature x and a target y?

A x is the model's prediction, while y is the algorithm.

B x represents information used to make the prediction, while y is the desired output.

C x is always numerical, while y is always categorical.

D x is used only during testing, while y is used only during training.

QUESTION 6

UNSUPERVISED LEARNING

A company has millions of customer records but no predefined customer categories. It wants an algorithm to automatically discover groups of customers with similar behavior. Which approach is most appropriate?

A Regression

B Classification

C Clustering

D Reinforcement learning

QUESTION 7

TRAINING PROCESS

During machine-learning training, the model repeatedly makes predictions, measures the error, and changes its parameters. What is the main purpose of this process?

A To increase the number of training samples

B To reduce the prediction error or training objective

C To automatically convert regression into classification

D To remove the need for features

QUESTION 8

GENERALIZATION

Two models are trained on the same dataset. Model A: Training accuracy = 99%, Test accuracy = 72%. Model B: Training accuracy = 91%, Test accuracy = 89%. Which statement is most reasonable?

A Model A generalizes better.

B Model B is likely to generalize better.

C Model B is definitely underfitting.

D Model A has lower variance.

QUESTION 9

MODEL CAPACITY

A regression model is too simple to capture the relationship between the input features and the target. Its predictions are inaccurate even for the training data. What would most likely help?

A. Use an even simpler model.

B. Reduce the amount of training data.

C. Increase the model's capacity or introduce more useful features.

D. Train only on the test dataset.

QUESTION 10

BIAS-VARIANCE BALANCE

Which statement best describes the ideal goal when choosing the complexity of a machine-learning model?

A Minimize training error regardless of test performance.

B Build the most complex possible model.

C Find a model that learns meaningful training patterns while generalizing well to unseen data.

D Ensure training accuracy is always 100%.

ANSWER KEY

Check your reasoning — not just the letter.

1	C	Continuous numerical target → regression.
2	B	Large train–test gap → overfitting.
3	B	Poor on train and test → underfitting.
4	C	Spam/not spam is binary classification.
5	B	x = input features; y = target output.
6	C	No labels + discover groups → clustering.
7	B	Training updates parameters to reduce loss.
8	B	Model B has the smaller generalization gap.
9	C	More capacity/useful features can reduce underfitting.
10	C	Goal: fit useful patterns and generalize to unseen data.

Key concepts: regression • overfitting • underfitting • generalization